

LCD MODULE SPECIFICATION

Model:	ALL-UE070WS-RB30-L106A
Version:	V1.0
Date:	2025.11.30

Customer Confirmation

Approved by	Notes

Please return one of the copies of the specification with your signature to us within two weeks after you receive this document. If it is not returned, we will assume that you agree to the entire contents of this specification document.

VIEWE Confirmation

Prepared by	Reviewed by	Approved by

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remarks
V1.0	2025.11.30	Preliminary release	All

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1.GENERAL INFORMATION

1.1 Features

- 1) Pixel Arrangement: RGB Vertical Stripe
- 2) Interface Mode: MIPI
- 3) Driver IC: JD9165BA ; Touch IC: GT911
- 4) Operation Temperature: -20~70°C
- 5) Storage Temperature: -30~80°C
- 6) Backlight Type: White LED
- 7) Display mode: Normally black
- 8) LED life time: 30,000 Hours

1.2 Mechanical Specification

Item 项目	Specification 规格	Unit 单位	Remark 备注
Pixel Driving element	TFT	-	-
Screen Size	7.0	Inch	Diagonal
Resolution	1024(RGB)X600	Dots	-
Active Area	154.21(H)X85.92 (V)	mm	-
Pixel pitch (W*H)	0.0502(H) X RGB X 0.1432(V)	mm	-
Module Size (W*H*D)	164.9 (H)X100.0(V)X4.83(T)	mm	-
Interface Mode	MIPI	-	-
Luminance	350	cd/m ²	Typ.
Viewing Direction	ALL	O'clock	-
Display Color	16.7M	Colors	24bits

2. ABSOLUTE MAXIMUM RATINGS

Item 项目	Symbol 符号	Min. 最小值.	Max. 最大值	Unit 单位	Remark 备注
Supply power voltage	VDD	-0.3	3.96	V	Note1
I/O power voltage	VDDIO	-0.3	3.96	V	Note1
AVDD input power	AVDD	-0.3	12	V	Note1
Gate On Power Supply Voltage	VGH	0.3	VGL+32	V	Note1
Gate Off Power Supply Voltage	VGL	VGH-32	0.3	V	Note1
VMID voltage	VMID	0.3	6.6	V	Note1
Common Power Supply Voltage	VCOM	0.3	5.43	V	Note1
VOTP (OTP power)	VOTP	-	9	V	Note1
Operating Temperature	TOPR	-20	-70	°C	Note1
Storage temperature	TSTG	-30	-80	°C	Note1

(Ta=+25°C,GND=0V)

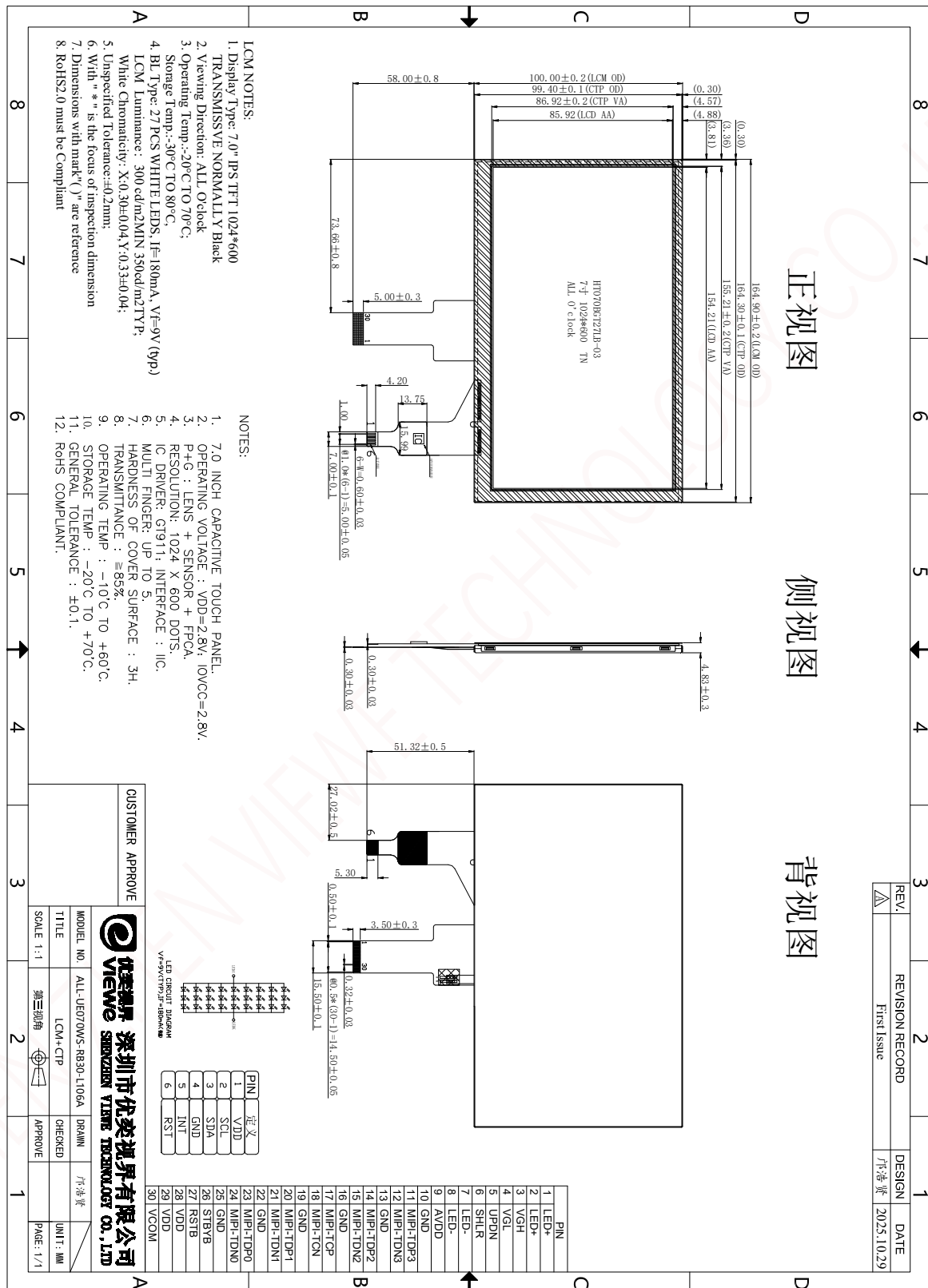
Note1:If the module exceeds the absolute maximum ratings, it may be damaged permanently. Also if the module operates with the absolute maximum ratings for a long time, the reliability may drop.

Note2: In case of temperature below 0°C,the response time of liquid crystal (LC) becomes slower and the color of panel darker than normal one.

Note3: Temp. ≤ 60°C , 90% RH MAX.

Temp. >60°C , Absolute humidity shall be less than 90% RH.

3. MECHANICAL DRAWING



4. I/O CONNECTION

PinNo.序号	Symbol符号	I/O
1	LED A	LED anode
2	LED A	LED anode
3	VGH	Gate ON Voltage
4	VGL	Gate OFF Voltage
5	UPDN	Vertical scan direction
6	SHLR	Horizontal scan direction
7	LED K	LED CATHODE
8	LED K	LED CATHODE
9	AVDD	Power for Analog Circuit
10	GND	Ground
11	MIPI_TDP3	MIPI differential data input
12	MIPI_TDN3	MIPI differential data input
13	GND	GND
14	MIPI_TDP2	MIPI differential data input
15	MIPI_TDN2	MIPI differential data input
16	GND	GND
17	MIPI_TCP	MIPI differential clock input
18	MIPI_TCN	MIPI differential clock input
19	GND	GND
20	MIPI_TDP1	MIPI differential data input
21	MIPI_TDN1	MIPI differential data input
22	GND	GND
23	MIPI_TDP0	MIPI differential data input
24	MIPI_TDN0	MIPI differential data input
25	GND	GND
26	STBYB	Standby Mode select pin.
27	RESET	Global reset pin
28	VDD	Power for Digital Circuit
29	VDD	Power for Digital Circuit
30	VCOM	Common-mode voltage

5. ELECTRICAL CHARACTERISTICS

5.1 TFT-LCD Panel Driving Section

Item 项目	Symbol 符号	Min. 最小值	Typ. 典型值	Max. 最大值	Unit 单位	Remark 备注
Power Supply Voltage	VDD	1.71	1.8	1.89	V	-
I/O power voltage	VDDIO	1.71	1.8	1.89	V	Note1
AVDD input voltage	AVDD	9	9.8	12	V	-
Gate on power supply	VGH	18	-	20	V	-
Gate off power supply	VGL	-8.5		-7.5	V	Note1
Common power supply	VCOM	2.7	3.6	5.4	V	Note1,2

(Ta=+25°C,GND=0V)

Note1:Measurement Conditions (Video Mode): Full Screen Red Pattern,60Hz Refresh.

5.2 Back Light Driving Section

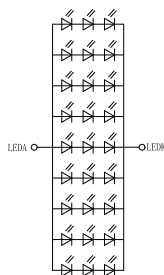
Item	Symbol	Min.	Typ.	Max.	Unit	Remark
Forward Voltage	V _F	8.1	9.0	9.9	V	Note1
Forward Current	I _F	-	180	-	mA	Note1
Backlight Power consumption	P _{BL}	-	1.620	-	Watt	Note1
LED Quantity		27			PCS	

(Ta=+25°C,GND=0V)

Note1: The LED driving condition is defined for each LED module.

For each LED : I_F=20mA,V_F=3V(Typ.)/3.3V(Max.),Ta=25°C。

Note2:The “LED life time” is defined as the module brightness decrease to 50% of original brightness at I_{LED}=20mA(Per Led). The LED life time could be decreased if operating I_{LED} is larger than 20mA.



LED CIRCUIT DIAGRAM
V_f=9V(TYP),I_F=180mA(每串)

5.3 Power On/Off Sequence

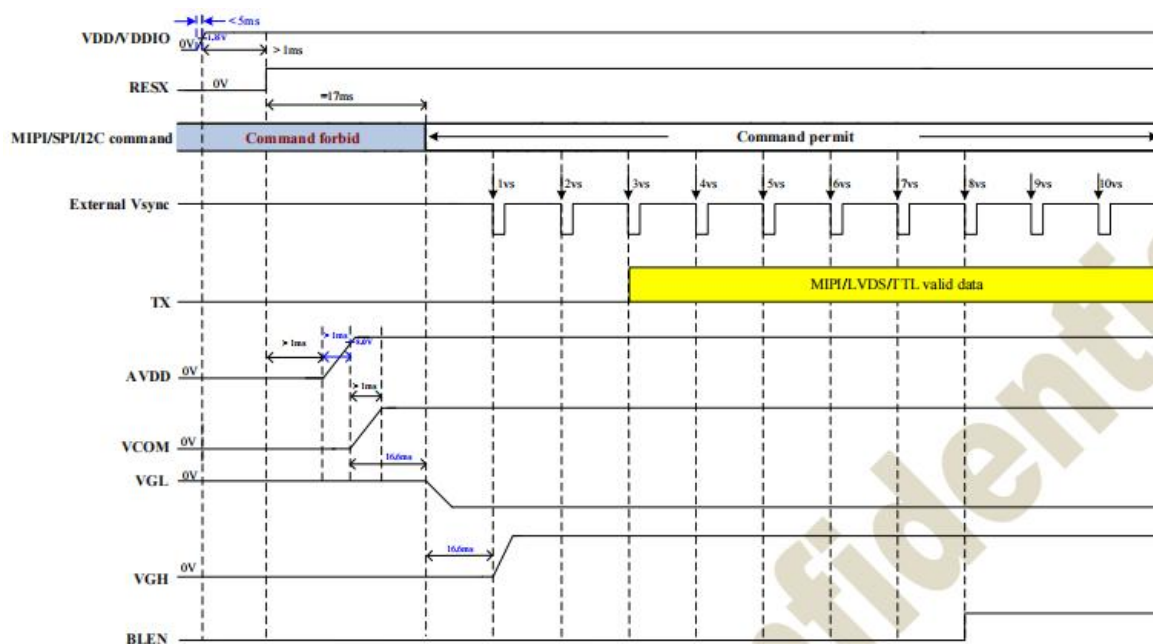


Figure 5.1: Power On timing chart

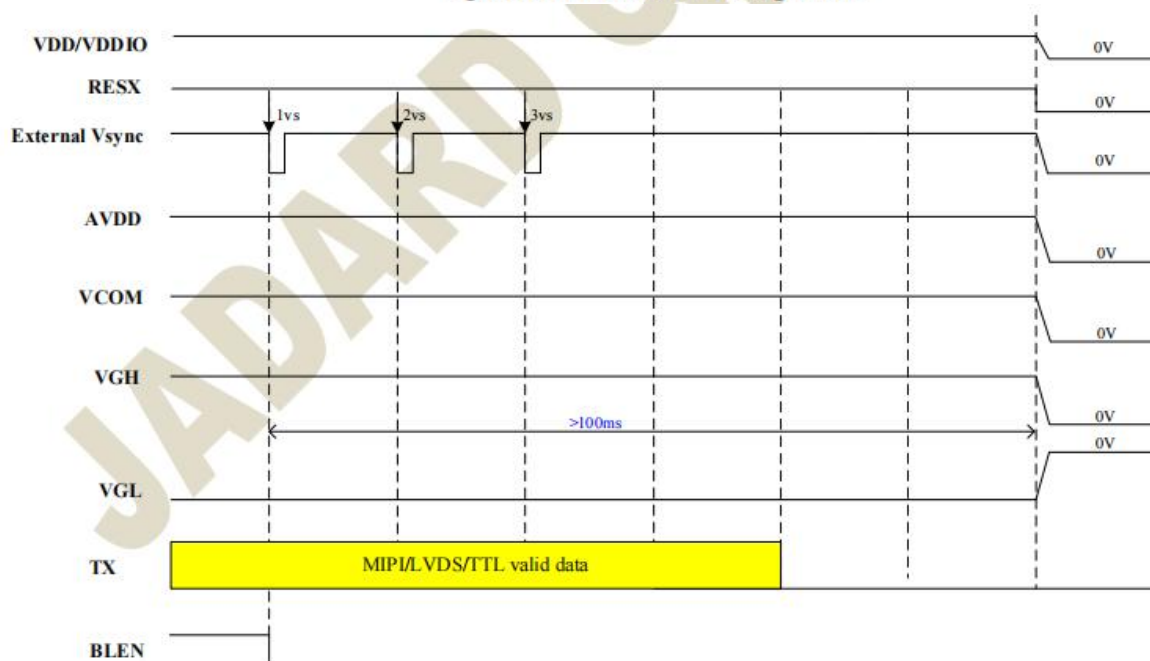


Figure 5.2: Power Off timing chart

5.4 Timing Characteristics

5.4.1 AC CHARACTERISTICS

Parameter	Descript	Spec.			Unit
		Min.	Typ.	Max.	
T_{REOT}	30%-85% rise time and fall time	-	-	35	ns
$T_{CLK-MISS}$	Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.	-	-	60	ns
$T_{CLK-POST}^{*1}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$.	$60\text{ ns} + 52*UI$	-	-	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8	-	-	UI
$T_{CLK-SETTLE}$	Time interval during which the HS receiver shall ignore any Clock Lane HS transitions, starting from the beginning of $T_{CLK-PRE}$.	95	-	300	ns
$T_{CLK-TERM-EN}$	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$.	Time for Dn to reach $V_{TERM-EN}$	-	38	ns
$T_{HS-SETTLE}$	Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of $T_{HS-PREPARE}$.	$85\text{ ns} + 6*UI$	-	$145\text{ ns} + 10*UI$	ns
T_{EOT}	Time from start of $T_{HS-TRAIL}$ or $T_{CLK-TRAIL}$ period to start of LP-11 state	-	-	$105\text{ ns} + n*12*UI$	-
$T_{HS-EXIT}^{(1)}$	time to drive LP-11 after HS burst	100	-	-	ns
$T_{HS-PREPARE}$	Time to drive LP-00 to prepare for HS transmission	$40\text{ ns} + 4*UI$	-	$85\text{ ns} + 6*UI$	ns
$T_{HS-PREPARE} + T_{HS-ZERO}$	$T_{HS-PREPARE}$ + Time to drive HS-0 before the Sync sequence	$145\text{ ns} + 10*UI$	-	-	ns
$T_{HS-SKIP}$	Time-out at RX to ignore transition period of EoT	40	-	$55\text{ ns} + 4*UI$	ns
$T_{HS-TRAIL}$	Time to drive flipped differential state after last payload data bit of a HS transmission burst	$60 + 4*UI$	-	-	ns
T_{LPX}	Length of any Low-Power state period	100	-	-	ns
Ratio T_{LPX}	Ratio of $T_{LPX(MASTER)}/T_{LPX(SLAVE)}$ between Master and Slave side	2/3	-	3/2	-
T_{TA-GET}	Time to drive LP-00 by new TX	$5*T_{LPX}$			ns
T_{TA-GO}	Time to drive LP-00 after Turnaround Request	$4*T_{LPX}$			ns
$T_{TA-SURE}$	Time-out before new TX side starts driving	T_{LPX}	-	$2*T_{LPX}$	ns

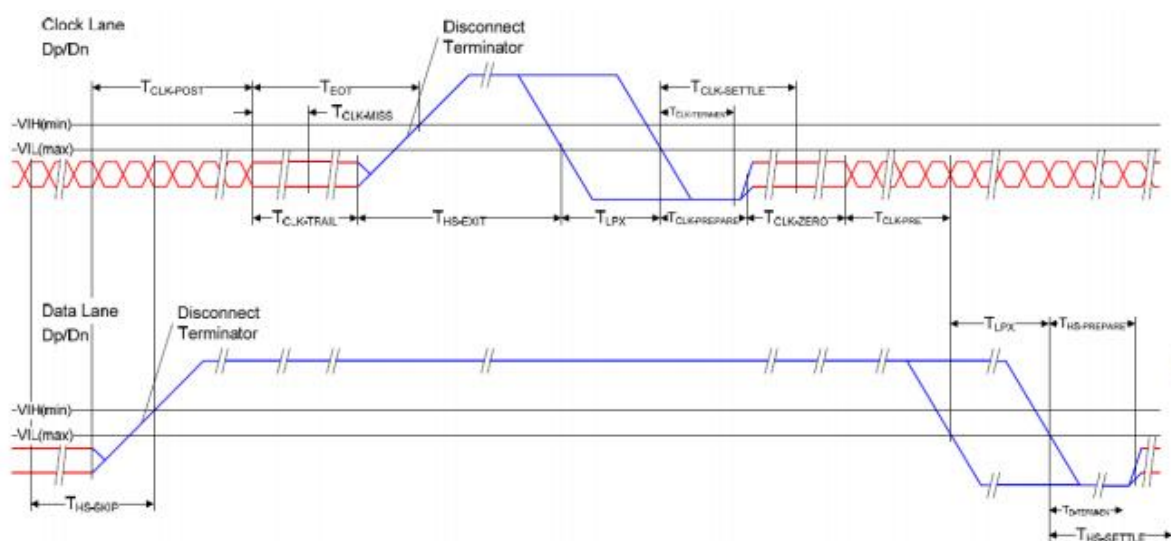


Figure 8.31: Switching the clock lane between clock transmission and low-po

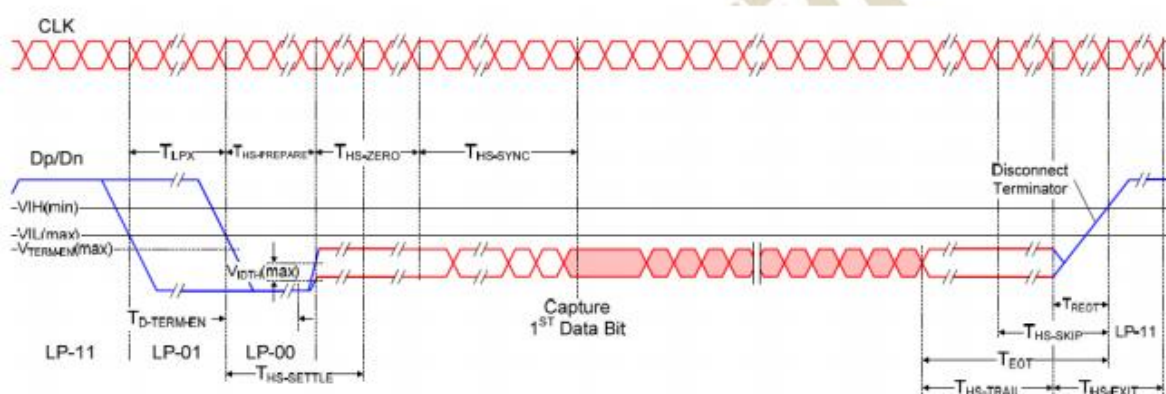
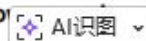


Figure 8.32: Timing of high-speed data transmission in bursts

5.4.2 DC CHARACTERISTICS

Parameter	Symbol	Spec.			Unit	Note
		Min.	Typ.	Max.		
Supply power voltage	VDD	-0.30	-	3.96	V	
IO supply voltage	VDDIO	-0.30		3.96	V	
AVDD voltage	AVDD	-0.30	-	12	V	
VGH voltage	VGH	-0.30	-	VGL+32V	V	
VGL voltage	VGL	VGH-32V	-	0.30	V	
VMID voltage	VMID	-0.30	-	6.6	V	
VCOM voltage	VCOM_OP	-0.30	-	5.43	V	
VOTP (OTP power)	VOTP	-	-	9	V	
Operating Temperature	T _{OPR}	-30	-	+85	°C	
Storage temperature	T _{STG}	-55	-	125	°C	

Parameter	Symbol	Spec.			Unit	Note
		Min.	Typ.	Max.		
Supply power voltage	VDD	1.71	1.8	1.89	V	MIPI mode condition1
I/O power voltage	VDDIO	1.71	1.8	1.89	V	
Supply power voltage	VDD	2.8	3.3	3.6	V	TTL mode condition LVDS mode condition MIPI mode condition2
I/O power voltage	VDDIO	2.8	3.3	3.6	V	
AVDD voltage	AVDD	9	-	12	V	
VGH voltage	VGH	15	18	20	V	
VGL voltage	VGL	-12	-8	-6	V	
VMID voltage	VMID	4.5	5	6.0	V	
VMID_OP voltage	VMID_OP	4.5	5	6.0	V	
VQHO voltage	VQHO	-	0.75AVDD	-	V	
VQLO voltage	VQLO	-	0.25AVDD	-	V	
VCOM voltage	VCOM_OP	2.7	3.8	5.4	V	
VOTP	VOTP	8.0	8.25	8.5	V	
Low level input voltage	V _{IL}	VSS	-	0.2xVDD	V	
High level input voltage	V _{IH}	0.8xVDD	-	VDD	V	
Low level output voltage	V _{OL}	VSS	-	0.1xVDD	V	
High level output voltage	V _{OH}	0.9xVDD	-	VDD	V	

6. OPTICAL CHARACTERISTICS

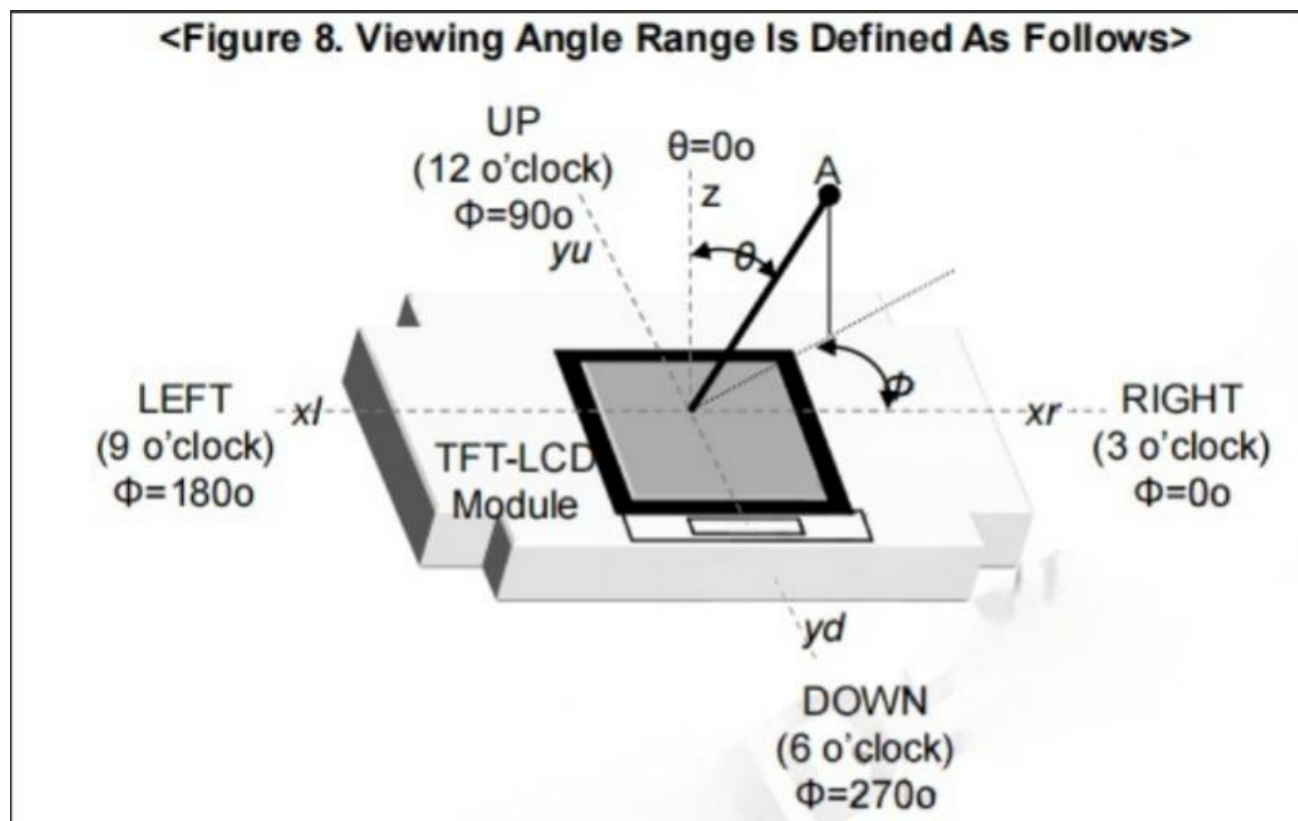
Parameter 参数	Symbol 符号	Condition 条件	Min. 最小值	Typ. 典型值	Max. 最大值	Unit 单位	Remark 备注
Contrast Ratio	C/R	$\theta = 0^{\circ}$	800	1000	-	-	APF Note 2
Luminance Uniformity(9P)	LU	-	75	-	-	%	
Response Time	T _r + T _F	25 °C	-	30	40	ms	Note 5
Color Coordination	W _X	$\theta = 0^{\circ}$ (Center) Normal viewing angle B/L On	+/-0.03	0.27	+/-0.03	-	With BLU @C Light Note 4
	W _Y			0.275			
	R _X		+/-0.03	Typ	+/-0.03		
	R _Y			Typ			
	G _X			Typ			
	G _Y			Typ			
	B _X			Typ			
	B _Y			Typ			
Viewing Angle	θ_{L}	C/R>10	-	85	-	Degree	Note 1
	θ_{R}		-	85	-		
	θ_{U}		-	85	-		
	θ_{D}		-	85	-		

Test Conditions:

VDD=3.3V, $I_F=20\text{mA}$ (Backlight current), the ambient temperature is +25°C.

Note 1: Viewing angle is the angle at which the contrast ratio is greater than 10. The viewing angles are determined for the horizontal or 3, 9 o'clock direction and the vertical or 6, 12

o'clock direction with respect to the optical axis which is normal to the LCD surface (see Figure 8).



Note 2: Contrast measurements shall be made at viewing angle of $\Theta=0^\circ$ and at the center of the LCD surface. Luminance shall be measured with all pixels in the view field set first to white, then to the dark (black) state. Luminance Contrast Ratio (CR) is defined mathematically.

$$CR = \frac{\text{Luminance when displaying a white raster}}{\text{Luminance when displaying a black raster}}$$

Note 3: Transmittance is the Value with Polarizer.

Note 4: The color chromaticity coordinates specified in Table 16 shall be calculated from the spectral data measured with all pixels first in red, green, blue and white. Measurements shall be made at the center of the panel.

Note 5: The electro-optical response time measurements shall be made as Figure 9 by switching the "data" input signal ON and OFF. The times needed for the luminance to change from 10% to 90% is T_r , and 90% to 10% is T_f .

7.RELIABILITY

Item 项目	Test Condition 测试条件	Remark 备注
High Temperature Storage	Ta =+70°C / 96Hours	Note1
Low Temperature Storage	Ta =-20°C / 96Hours	
High Temperature Operating	Ta =+80°C / 96Hours	
Low Temperature Operating	Ta =-30°C / 96Hours	
Temperature Cycle storage Test	-10-60'C,100cycles, 1hr/cycle.Non-operational	
Thermal humidity storage Test	60°C x 90%RH / 240Hours	Note2
Package Vibration Test	Frequency: 10Hz~55Hz,Amplitude:1.5mm, 1 hrs for each direction of X, Y, Z	
Packing shock test	Drop to the ground from 60cm height, 1 corner, 3 edges, 6 surfaces.	

Inspection after Test :

Note1:The test result shall be evaluated after the sample has been left at room temperature and humidity for 1 hours.

These defects can't be accepted:1.Air bubble 2.Seal leak 3.Non-display 4.Missing segments 5.Glass crack. 6. The reliability test result is based on XXXX, XXXX, XXXX, XXXX IC

Note 2: In the standard condition, there shall be no practical problem that may affect the display function. After the reliability test, the product only guarantees operation, but doesn't guarantee all the cosmetic specification.

8.Specification of Quality Assurance:

8-1. Purpose

This standard for Quality Assurance should affirm the quality of LCD module products to supply to purchaser by VIEWE.

8-2. Standard for Quality Test

a. Inspection:

Before delivering, the supplier should take the following tests, and affirm the quality of product.

b. Electro-Optical Characteristics:

According to the individual specification to test the product.

c. Test of Appearance Characteristics:

According to the individual specification to test the product.

d. Test of Reliability Characteristics:

According to the definition of reliability on the specification for testing products.

e. Delivery Test :

Before delivering, the supplier should take the delivery test.

(i) Test method: According to ISO2859-1. General Inspection Level II take a single time.

(ii) The defects classify of AQL as following :

Major defect: AQL = 0.65

Minor defect: AQL = 2.5

Total defects: AQL = 2.5

8-3. Non- conforming Analysis & Deal With Manners

a. Non- conforming Analysis :

(i) Purchaser should supply the detail data of non- conforming sample and the non- conforming.

(ii) After accepting the detail data from purchaser, the analysis of non- conforming should be finished in two weeks.

(iii) If supplier can not finish analysis on time, must announce purchaser before 3 days.

b. Disposition of non- conforming:

(i) If find any product defect of supplier during assembly time, supplier must change the good product for every defect after recognition.

(ii) Both supplier and customer should analyze the reason and discuss the disposition of non- conforming when the reason of nonconforming is not sure.

8-4. Agreement items

Both sides should discuss together when the following problems happen.

a. There is any problem of standard of quality assurance, and both sides should think that must be modified.

b. There is any argument item which does not record in the standard of quality assurance.

c. Any other special problem.

8-5. Standard of The Product Appearance Test

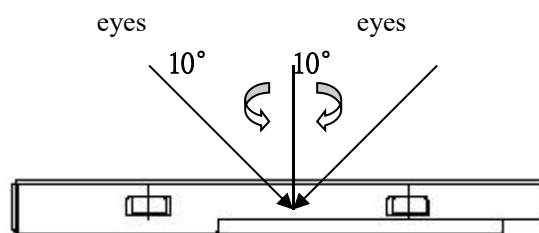
a. Manner of appearance test:

(i) The test must be under $20W \times 2$ or $40W$ fluorescent light, and the distance of view must be at $30 \pm 5cm$.

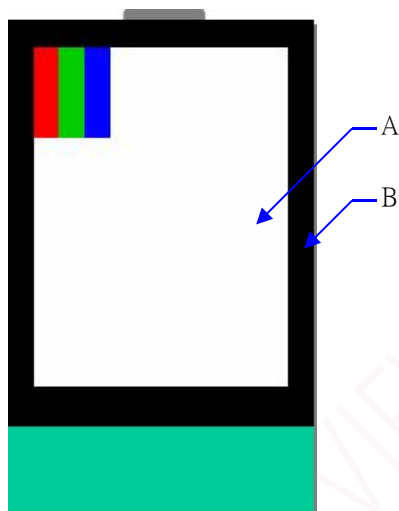
(ii) When test the model of transmissive product must add the reflective plate.

(iii) The test direction is base on around 10° of vertical line.

(iii) Temperature: $25 \pm 5^\circ C$ Humidity: $60 \pm 10\% RH$



(iv) Definition of area :



A. Area: Viewing area.

B. Area: Out of viewing area.

(Outside viewing area)

b. Basic principle::

(i) It will accord to the AQL when the standard can not be described.

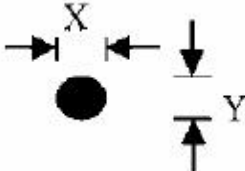
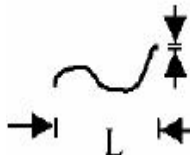
(ii) The sample of the lowest acceptable quality level must be discussed by both supplier and customer when any dispute happened.

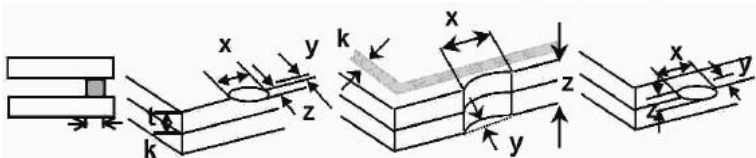
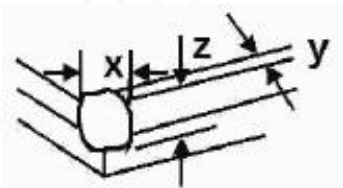
(iii) Must add new item on time when it is necessary.

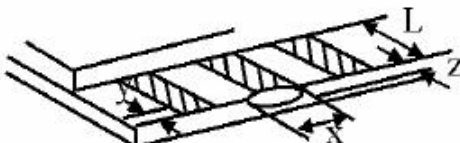
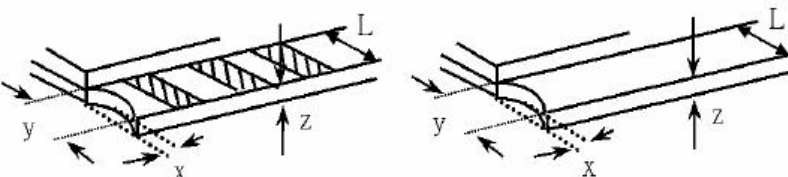
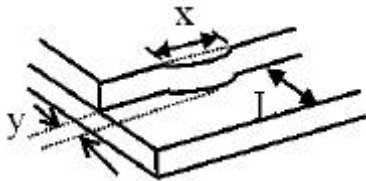
c. Standard of inspection : (Unit: mm)

8-6. Inspection specification

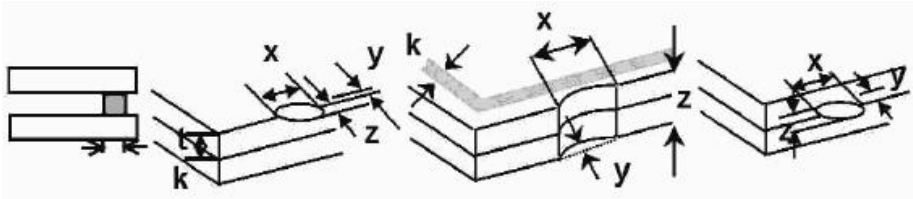
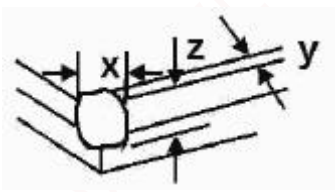
Defect out of viewing area can be neglected

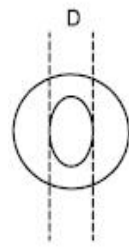
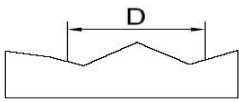
NO	Item	Criterion	AQL																										
01	Electrical Testing	1.1 Missing vertical, horizontal segment, segment contrast defect. 1.2 Missing character, dot or icon. 1.3 Display malfunction. 1.4 No function or no display. 1.5 Current consumption exceeds product specifications. 1.6 LCD viewing angle defect. 1.7 Mixed product types. 1.8 Flicker	0.65																										
02	Black or White spots or Bright spots or Color spots on LCD (Display only)	2.1 White and black or color spots on display $\leq 0.25\text{mm}$, no more than Five spots. 2.2 Densely spaced: No more than three spots within 3mm.	2.5																										
03	LCD and Touch Panel black spots, white spots, contamination (non – display)	3.1 Round type: As following drawing $\Phi = (X+Y) / 2$  <table border="1"> <thead> <tr> <th>Size(mm)</th> <th>Acceptable Q'ty</th> </tr> </thead> <tbody> <tr> <td>$\Phi \leq 0.10$</td> <td>Accept no dense</td> </tr> <tr> <td>$0.10 < \Phi \leq 0.20$</td> <td>3</td> </tr> <tr> <td>$0.20 < \Phi \leq 0.25$</td> <td>2</td> </tr> <tr> <td>$0.25 < \Phi \leq 0.30$</td> <td>1</td> </tr> <tr> <td>$0.30 < \Phi$</td> <td>0</td> </tr> </tbody> </table> * Densely spaced: No more than two spots within 3mm. 3.2 Line type: (As following drawing)  <table border="1"> <thead> <tr> <th>Length(mm)</th> <th>Width(mm)</th> <th>Acceptable Q'ty</th> </tr> </thead> <tbody> <tr> <td>---</td> <td>$W \leq 0.02$</td> <td>Accept no dense</td> </tr> <tr> <td>$L \leq 3.0$</td> <td>$0.02 < W \leq 0.05$</td> <td rowspan="2">2</td> </tr> <tr> <td>$L \leq 2.5$</td> <td>$0.03 < W \leq 0.15$</td> </tr> <tr> <td>---</td> <td>$0.15 < W$</td> <td>Rejection</td> </tr> </tbody> </table> * Densely spaced: No more than two lines within 3mm.	Size(mm)	Acceptable Q'ty	$\Phi \leq 0.10$	Accept no dense	$0.10 < \Phi \leq 0.20$	3	$0.20 < \Phi \leq 0.25$	2	$0.25 < \Phi \leq 0.30$	1	$0.30 < \Phi$	0	Length(mm)	Width(mm)	Acceptable Q'ty	---	$W \leq 0.02$	Accept no dense	$L \leq 3.0$	$0.02 < W \leq 0.05$	2	$L \leq 2.5$	$0.03 < W \leq 0.15$	---	$0.15 < W$	Rejection	2.5
Size(mm)	Acceptable Q'ty																												
$\Phi \leq 0.10$	Accept no dense																												
$0.10 < \Phi \leq 0.20$	3																												
$0.20 < \Phi \leq 0.25$	2																												
$0.25 < \Phi \leq 0.30$	1																												
$0.30 < \Phi$	0																												
Length(mm)	Width(mm)	Acceptable Q'ty																											
---	$W \leq 0.02$	Accept no dense																											
$L \leq 3.0$	$0.02 < W \leq 0.05$	2																											
$L \leq 2.5$	$0.03 < W \leq 0.15$																												
---	$0.15 < W$	Rejection																											

NO	Item	Criterion		AQL																		
04	Polarizer bubbles	If bubbles are visible, judge using black spot specifications, not easy to find, must check in specify direction	<table><tr><th>Size Φ(mm)</th><th>Acceptable Q'ty</th></tr><tr><td>$\Phi \leq 0.20$</td><td>Accept no dense</td></tr><tr><td>$0.20 < \Phi \leq 0.50$</td><td>3</td></tr><tr><td>$0.50 < \Phi \leq 1.00$</td><td>2</td></tr><tr><td>$1.00 < \Phi$</td><td>0</td></tr><tr><td>Total Q'ty</td><td>3</td></tr></table>	Size Φ (mm)	Acceptable Q'ty	$\Phi \leq 0.20$	Accept no dense	$0.20 < \Phi \leq 0.50$	3	$0.50 < \Phi \leq 1.00$	2	$1.00 < \Phi$	0	Total Q'ty	3	2.5						
Size Φ (mm)	Acceptable Q'ty																					
$\Phi \leq 0.20$	Accept no dense																					
$0.20 < \Phi \leq 0.50$	3																					
$0.50 < \Phi \leq 1.00$	2																					
$1.00 < \Phi$	0																					
Total Q'ty	3																					
05	Scratches	Follow NO.3 -2 Line Type.																				
06	Chipped glass	Symbols: x: Chip length y: Chip width z: Chip thickness k: Seal width t: Glass thickness a: LCD side length L: Electrode pad length 6.1 General glass chip: 6.1.1 Chip on panel surface and crack between panels :  <table><tr><td>z: Chip thickness</td><td>y: Chip width</td><td>x: Chip length</td></tr><tr><td>$Z \leq 1/2t$</td><td>Not over viewing are</td><td>$x \leq 1/8a$</td></tr><tr><td>$1/2t < z \leq 2t$</td><td>Not exceed 1/3k</td><td>$x \leq 1/8a$</td></tr></table> ⊙ Unit: mm ⊙ If there are 2 or more chips, x is the total length of each chip 6.1.2 Corner crack :  <table><tr><td>z: Chip thickness</td><td>y: Chip width</td><td>x: Chip length</td></tr><tr><td>$Z \leq 1/2t$</td><td>Not over viewing area</td><td>$x \leq 1/8a$</td></tr><tr><td>$1/2t < z \leq 2t$</td><td>Not exceed 1/3k</td><td>$x \leq 1/8a$</td></tr></table> ⊙ Unit: mm ⊙ If there are 2 or more chips, x is the total length of each chip		z: Chip thickness	y: Chip width	x: Chip length	$Z \leq 1/2t$	Not over viewing are	$x \leq 1/8a$	$1/2t < z \leq 2t$	Not exceed 1/3k	$x \leq 1/8a$	z: Chip thickness	y: Chip width	x: Chip length	$Z \leq 1/2t$	Not over viewing area	$x \leq 1/8a$	$1/2t < z \leq 2t$	Not exceed 1/3k	$x \leq 1/8a$	2.5
z: Chip thickness	y: Chip width	x: Chip length																				
$Z \leq 1/2t$	Not over viewing are	$x \leq 1/8a$																				
$1/2t < z \leq 2t$	Not exceed 1/3k	$x \leq 1/8a$																				
z: Chip thickness	y: Chip width	x: Chip length																				
$Z \leq 1/2t$	Not over viewing area	$x \leq 1/8a$																				
$1/2t < z \leq 2t$	Not exceed 1/3k	$x \leq 1/8a$																				

NO	Item	Criterion	AQL																
07	Glass crack	Symbols: x: Chip length y: Chip width z: Chip thickness k: Seal width t: Glass thickness a: LCD side length L: Electrode pad length 7.2 Protrusion over terminal : 7.2.1 Chip on electrode pad :  <table><tr><td>y: Chip width</td><td>x: Chip length</td><td>z: Chip thickness</td></tr><tr><td>$y \leq 0.5\text{mm}$</td><td>$x \leq 1/8a$</td><td>$0 < z \leq t$</td></tr></table> 7.2.2 Non-conductive portion :  <table><tr><td>y: Chip width</td><td>x: Chip length</td><td>z: Chip thickness</td></tr><tr><td>$y \leq L$</td><td>$x \leq 1/8a$</td><td>$0 < z \leq t$</td></tr></table> ⊙ If there chipped area touches the ITO terminal, over 2/3 of the ITO must remain and be inspected according to electrode terminal specifications. ⊙ If the product will be heat sealed by the customer, the alignment mark must not be damaged. 7.2.3 Substrate protuberance and internal crack  <table><tr><td>y: width</td><td>x: length</td></tr><tr><td>$y \leq 1/3L$</td><td>$X \leq a$</td></tr></table>	y: Chip width	x: Chip length	z: Chip thickness	$y \leq 0.5\text{mm}$	$x \leq 1/8a$	$0 < z \leq t$	y: Chip width	x: Chip length	z: Chip thickness	$y \leq L$	$x \leq 1/8a$	$0 < z \leq t$	y: width	x: length	$y \leq 1/3L$	$X \leq a$	2.5
		y: Chip width	x: Chip length	z: Chip thickness															
		$y \leq 0.5\text{mm}$	$x \leq 1/8a$	$0 < z \leq t$															
y: Chip width	x: Chip length	z: Chip thickness																	
$y \leq L$	$x \leq 1/8a$	$0 < z \leq t$																	
y: width	x: length																		
$y \leq 1/3L$	$X \leq a$																		

NO	Item	Criterion	AQL
08	Cracked glass	The LCD with extensive crack is not acceptable.	2.5
09	Backlight elements	9.1 Illumination source flickers when lit. 9.2 Spots or scratches that appear when lit must be judged. Using LCD spot, lines and contamination standards. 9.3 Backlight doesn't light or color is wrong.	2.5 2.5 0.65
10	Bezel	Bezel must comply with product specifications.	2.5
11	PCB、COB	11.1 COB seal may not have pinholes larger than 0.2mm or contamination. 11.2 COB seal surface may not have pinholes through to the IC. 11.3 The height of the COB should not exceed the height indicated in the assembly diagram. 11.4 There may not be more than 2mm of sealant outside the seal area on PCB. And there should be no more than three places. 11.5 Parts on PCB must be the same as on the production characteristic chart, There should be no wrong parts, missing parts or excess parts. 11.6 The jumper on the PCB should conform to the product characteristic chart.	2.5 2.5 2.5 2.5 0.65 0.65
12	FPC	12.1 FPC terminal damage $\leq$ 1/2 FPC terminal width and can not affect the function, we judge accept. 12.2 FPC alignment hole damage $\leq$ 1/2 alignment area and can not affect the function, we judge accept.	2.5 2.5
13	Soldering	13.1 No cold solder joints, missing solder connections, oxidation or icicle. 13.2 No short circuits in components on PCB or FPC.	2.5 0.65

NO	Item	Criterion	AQL												
14	Touch Panel Chipped glass	Symbols : x: Chip length y: Chip width z: Chip thickness k: Seal width t: Glass thickness a: LCD side length L: Electrode pad length 14.1 General glass chip : 14.1.1 Chip on panel surface and crack between panels:  <table> <tr> <td>z: Chip thickness</td> <td>y: Chip width</td> <td>x: Chip length</td> </tr> <tr> <td>$Z \leq t$</td> <td>$\leq 1/2 k$ and not over viewing area</td> <td>$x \leq 1/8a$</td> </tr> </table> ⊙ Unit: mm ⊙ If there are 2 or more chips, x is the total length of each chip 14.1.2 Corner crack:  <table> <tr> <td>z: Chip thickness</td> <td>y: Chip width</td> <td>x: Chip length</td> </tr> <tr> <td>$z \leq t$</td> <td>$\leq 1/2 k$ and not over viewing area</td> <td>$x \leq 1/8a$</td> </tr> </table> ⊙ Unit: mm ⊙ If there are 2 or more chips, x is the total length of each chip	z: Chip thickness	y: Chip width	x: Chip length	$Z \leq t$	$\leq 1/2 k$ and not over viewing area	$x \leq 1/8a$	z: Chip thickness	y: Chip width	x: Chip length	$z \leq t$	$\leq 1/2 k$ and not over viewing area	$x \leq 1/8a$	2.5
z: Chip thickness	y: Chip width	x: Chip length													
$Z \leq t$	$\leq 1/2 k$ and not over viewing area	$x \leq 1/8a$													
z: Chip thickness	y: Chip width	x: Chip length													
$z \leq t$	$\leq 1/2 k$ and not over viewing area	$x \leq 1/8a$													

NO	Item	Criterion	AQL										
15	Touch Panel(Fish eye、dent and bubble on film)	<table><tr><th>SIZE(mm)</th><th>Acceptable Q'ty</th></tr><tr><td>$\Phi \leq 0.2$</td><td>Accept no dese</td></tr><tr><td>$0.2 < D \leq 0.4$</td><td>5</td></tr><tr><td>$0.4 < D \leq 0.5$</td><td>2</td></tr><tr><td>$0.5 < D$</td><td>0</td></tr></table> 	SIZE(mm)	Acceptable Q'ty	$\Phi \leq 0.2$	Accept no dese	$0.2 < D \leq 0.4$	5	$0.4 < D \leq 0.5$	2	$0.5 < D$	0	2.5
SIZE(mm)	Acceptable Q'ty												
$\Phi \leq 0.2$	Accept no dese												
$0.2 < D \leq 0.4$	5												
$0.4 < D \leq 0.5$	2												
$0.5 < D$	0												
16	Touch Panel Newton ring	Newton ring dimension $\leq 1/2$ touch panel area and not affect font and line distortion ($\leq 2.5\%$), it is acceptable.	2.5										
17	Touch Panel Linearity	Less than 2.5% is acceptable.	2.5										
18	LCD Ripple	Touch the touch panel, can not see the LCD ripple. Pen: R 1.0mm silicon rubber. Operation Force: 80g	2.5										
19	General appearance	19.1 Pin type must match type in specification sheet. 19.2 LCD pin loose or missing pins. 19.3 Product packaging must the same as specified on packaging specification sheet. 19.4 Product dimension and structure must conform to product specification sheet.	0.65 0.65 0.65 0.65										

9. Handling Precaution :

9-1 Handling of LCM

- Don't give external shock.
- Don't apply excessive force on the surface.
- Liquid in LCD is hazardous substance. Must not lick and swallow. when the liquid is attach to your hand, skin, cloth etc. Wash it out thoroughly and immediately.
- Don't operate it above the absolute maximum rating.
- Don't disassemble the LCM.
- The operators should be grounded whenever he/she comes into contact with the module. Never touch any of the conductive parts such as the LSI pads, the copper leads on the PCB and the interface terminals with any parts of the human body.
- The modules should be kept in antistatic bags or other containers resistant to static for storage.
- The module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

9-2 Storage

- Store in an ambient temperature of $25\pm 10^{\circ}\text{C}$, and in a relative humidity of $50\pm 10\%\text{RH}$. Don't expose to sunlight or fluorescent light.
- Storage in a clean environment, free from dust, active gas, and solvent.
- Store in anti-static electricity container.
- Store without any physical load.

9-3 Soldering

- Use only soldering irons with proper grounding and no leakage.
- Iron: No higher than $280\pm 10^{\circ}\text{C}$ and less than 3 sec during Hand soldering.
- Rewiring: no more than 2 times.

10.PACKAGE DRAWING



第一步

将产品放入吸塑盘中，
LCD AA 面朝上，注意
防呆方向

第二步

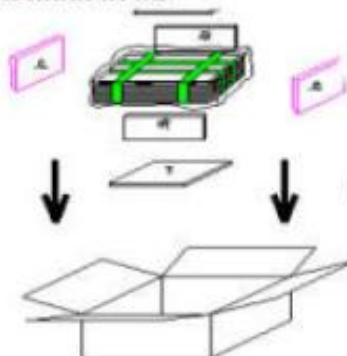
每一层吸塑盘与相邻
层，叠放时相错 180
度，最上层不放产品，
总叠加层数参考

第三步

检查无误后用胶带固
定吸塑盘，将捆好的吸
塑盘放入无尘带中并
封口；

First step

Putting products into the
tray,
LCD A.A faces Upward,
(pay attention to the



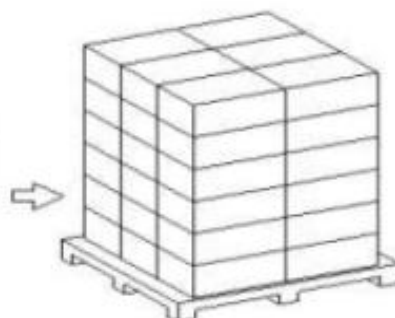
Second step

Neighbouring trays should be
staggered 180° while stacking
up.
on the top, there is an empty
tray without product



Third step

Taping up the tray
after inspection, and
put them into a PE



第四步

外箱内侧底部和四周
放上泡棉将包好的产品
装入纸箱，合盖；

第五步

最后胶带封箱，贴外箱
标签

第六步

将每箱整齐放在栈板
上并包裹最高可堆叠 6
层)

Fourth step

Putting EPE foams and
products with trays into
the carton;
Close the carton box

fifth step

Sealing the carton with
cellulose tape ;
Stick on a carton label,

sixth step

Placing the boxes together
on a pallet (6 layers at
most),